

Name of Department:- Computer Science and Engineering

1. Subject Code: Course Title:
2. Contact Hours: L: T: P:
3. Semester: VI
4. Pre-requisite: TCS 505
5. Course Outcomes: After completion of the course students will be able to

- 1.Characterize and appreciate computer networks from the view point of components and from the view point of services
2. Display good understanding of the flow of a protocol in general and a network protocol in particular
3. Model a problem or situation in terms of layering concept and map it to the TCP/IP stack
4. Select the most suitable Application Layer protocol (such as HTTP, FTP, SMTP, DNS, Bittorrent) as per the requirements of the network application and work with available tools to demonstrate the working of these protocols.
5. Design a Reliable Data Transfer Protocol and incrementally develop solutions for the requirements of Transport Layer
6. Describe the essential principles of Network Layers and use IP addressing to create subnets for any specific requirements

6. Detailed Syllabus

UNIT	CONTENTS	Contact Hrs
Unit – I	Introduction: Computer Networks and the Internet, Overall view: As components and as services; What is a protocol, what is a network protocol, Access Networks and Physical Media, Circuit and Packet Switching, Internet Backbone, Delays: Processing, Queuing, Transmission and Propagation delays The Layered Architecture: Protocol Layering, The OSI Reference Model and the TCP/IP protocol stack, History of Computer Networking and the Internet	11
Unit - II	Application Layer: Principles and Architectures of Network Applications, Client and Server processes, the idea of socket, Transport services available to Application Layer especially in the internet. Application Layer Protocols: The Web and http: Persistent and Non-persistent connections, http message format, cookies, proxy server, conditional GET File Transfer Protocol Email: smtp, mail message formats, mail access protocols: pop3, imap, MIME DNS: Services, How it works, Root, Top-Level and Authoritative DNS servers, Resource Records, DNS messages A simple introduction to p2p file distribution: BitTorrent	12

Unit – III	Transport Layer: Introduction and Services, The Transport layer in internet, Difference between Connection Oriented and Connectionless services UDP: Segment structure, checksum in UDP	6
Unit – IV	Transport Layer2: The principles behind connection oriented data transfer, designing a connection oriented protocol, stop-and-wait, Go Back N, Selective Repeat TCP: Connection Establishment, TCP header, Sequence and acknowledgement numbers, Round Trip Time, Flow Control, Congestion Control	6
Unit – V	Network Layer I: Introduction, Packet Forwarding and Routing, Difference between Virtual Circuits and Datagram networks, The internals of a router: Input ports, output ports, switching architecture The Internet Protocol(IP): Datagram format, IP fragmentation, IPv4 addressing, subnets, CIDR, classful addressing, DHCP, Network Address Translation(NAT), Universal Plug and Play as a provider of NAT, Internet Control Message Protocol(ICMP), IPv6 Header, Moving from IPv4 to IPv6: tunnelling, A brief discussion on IP security (Note: Network Layer will continue with Routing Algorithms in Computer Networks II in the next semester)	10
Total		45

Text Books:

1. Computer Networking: "A Top Down Approach (5th edition)", Ross and Kurose, Pearson/Addison-Wesley

Reference Books:

1. Andrew Tanenbaum and David Wetherhall, "Computer Networks(5th edition)", Prentice Hall
2. Peterson and Davie, "Computer Networks: A System Approach (4th edition)", Elsevier
3. Forouzan, "Data Communication and Networking (4th edition)", McGraw Hill
4. William Stallings: "Data and Computer Communication", 8th Edition, Pearson Education, 2007
5. Nader F. Mir: "Computer and Communication Networks", Pearson Education, 2007.

Name of Department:- Computer Science and Engineering

1. Subject Code: Course Title:

2. Contact Hours: L: T: P:

3. Semester: V

4. Pre-requisite: TCS 301, TCS 403, TCS 404

5. Course Outcomes: After completion of the course students will be able to

1. Understand the concept and design issues associated with an operating system
2. Identify the problems related to process management and synchronization and apply learned methods to solve basic problems
3. Explain the basics of memory management and the use of virtual memory in modern operating systems.
4. Understand the concept deadlock avoidance, prevention and detection techniques.
5. Implementation of process management, memory management and file management using system calls.
6. Analyze the data structures and algorithms used for developing an operating systems

6. Detailed Syllabus

UNIT	CONTENTS	Contact Hrs
Unit – I	Introduction to Operating Systems, UNIX: What operating systems do; Operating System structure; Operating System Services; User - Operating System interface; System calls; Types of system calls; System programs; Operating System structure; Unix command: Command Structure, Internal and External commands, filters; vi editor.	8
Unit - II	Process Management: Process concept; Process scheduling; Operations on processes; Multi-Threaded Programming: Overview; Multithreading models; Threading issues. Process Scheduling: Basic concepts; Scheduling criteria; Scheduling algorithms; Multiple-Processor scheduling; Thread scheduling. Process Synchronization: Inter-process communication; Synchronization: The Critical section problem; Peterson's solution; Synchronization hardware; Semaphores; Classical problems of synchronization.	10
Unit – III	Deadlocks: Deadlocks: System model; Deadlock characterization; Methods for handling deadlocks; Deadlock prevention; Deadlock avoidance; Deadlock detection and recovery from deadlock. Memory Management: Memory Management Strategies: Background; Swapping; Contiguous memory allocation; Paging; Structure of page table; Segmentation. Virtual Memory Management: Background; Demand paging; Page replacement; Allocation of frames; Thrashing.	10

Unit – IV	File System, Implementation of File System: File System: File concept; Access methods; Directory structure; File system mounting; File sharing; Protection. Implementing File System: File system structure; Directory implementation; Allocation methods; Free space management. Secondary Storage Structures, Protection : Mass storage structures; Disk structure; Disk scheduling; Disk management; Swap space management. Protection: Goals of protection, Principles of protection, Access matrix.	8
Unit – V	Shell Programming: Shell scripts, Running script in the current shell, Pattern Matching, Redirection, String handling, Conditional Parameter Substitution, Shell functions. Case Study: The Linux Operating System: Linux history; Design principles; Kernel modules; Process management; Scheduling; Memory management; File systems, Input and output; Inter-process communication.	8
Total		44

Text Books:

1. Abraham Silberschatz, Peter Baer Galvin, Greg Gagne: "Operating System Principles", 7th edition, Wiley India, 2006.
2. William Stallings: "Operating Systems: Internals and Design Principles", 6th edition, Pearson, 2009
3. Sumitabha Das, "Unix concepts and applications"

Reference Books:

1. Andrew S Tanenbaum: "Operating Systems: Design and Implementation", 3rd edition, Prentice Hall, 2006
2. Stuart E. Madnick, John Donovan: "Operating Systems", Tata McGraw Hill, 2008

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1. Subject Code: Course Title:
2. Contact Hours: L: T: P:
3. Semester: V
4. Pre-requisite: TCS 302, TCS 404

5. Course Outcomes: After completion of the course students will be able to

- Understand the different issues involved in the design and implementation of a database system.
- Study the physical and logical database designs, database modeling, relational, hierarchical, and network models
- Understand and use data manipulation language to query, update, and manage a database
- Develop an understanding of essential DBMS concepts such as: database security, integrity, concurrency,
- Design and build a simple database system and demonstrate competence with the fundamental tasks involved with modeling, designing, and implementing a DBMS.
- Evaluate a business situation and designing & building a database applications

6. Detailed Syllabus

UNIT	CONTENTS	Contact Hrs
Unit – I	Introduction: An overview of DBMS; Advantages of using DBMS approach; Database systems vs File Systems, Database system concepts and architecture Data models, schemas and instances; Three-schema architecture and data independence; Database languages and interfaces; The database system environment; Centralized and client-server architectures; Classification of Database Management systems.	9
Unit – II	Entity-Relationship Model: Using High-Level Conceptual Data Models for Database Design; An Example Database Application; Entity Types, Entity Sets, Attributes and Keys; Relationship types, Relationship Sets, Roles and Structural Constraints; Weak Entity Types; Refining the ER Design; ER Diagrams, Naming Conventions and Design Issues; Relationship types of degree higher than two.	9
Unit – III	Relational Model and Relational Algebra : Relational Model Concepts; Relational Model Constraints and Relational Database Schemas; Update Operations, Transactions and dealing with constraint violations; Unary Relational Operations: SELECT and PROJECT; Relational Algebra Operations from Set Theory; Binary Relational Operations : JOIN and DIVISION; Additional Relational Operations;	11

	Examples of Queries in Relational Algebra; Relational Database Design Using ER- to-Relational Mapping. SQL – 1: SQL Data Definition and Data Types; Specifying basic constraints in SQL; Schema change statements in SQL; Basic queries in SQL; More complex SQL Queries.	
Unit – IV	Insert, Delete and Update statements in SQL; Specifying constraints as Assertion and Trigger; Views (Virtual Tables) in SQL; Additional features of SQL; Database programming issues and techniques; Embedded SQL, Dynamic SQL; Database stored procedures.	9
Unit – V	Database Design – 1: Informal Design Guidelines for Relation Schemas; Functional Dependencies; Normal Forms Based on Primary Keys; General Definitions of Second and Third Normal Forms; Boyce-Codd Normal Form Properties of Relational Decompositions; Algorithms for Relational Database Schema Design; Multivalued Dependencies and Fourth Normal Form; Join Dependencies and Fifth Normal Form; Inclusion Dependencies; Other Dependencies and Normal Forms Transaction Management: The ACID Properties; Transactions and Schedules; Concurrent Execution of Transactions; Lock- Based Concurrency Control; Performance of locking; Transaction support in SQL; Introduction to crash recovery; 2PL, Serializability and Recoverability; Lock Management; Log Files; Checkpointing; Recovering from a System Crash; Media Recovery	9
Total		47

Text Books:

- Elmasri and Navathe: "Fundamentals of Database Systems", 5th Edition, Pearson Education, 2007.
- Raghu Ramakrishnan and Johannes Gehrke: " Database Management Systems", 3rd Edition, McGraw-Hill, 2003.

Name of Department:- Computer Science and Engineering

1. Subject Code: Course Title:
2. Contact Hours: L: T: P:
3. Semester: V
4. Pre-requisite: TCS201, TCS421

5. Course Outcomes: After completion of the course students will be able to

1. Have a good understanding of the fundamental issues and challenges of machine learning: data, model selection, model complexity, etc.
2. Have an understanding of the strengths and weaknesses of many popular machine learning approaches.
3. Appreciate the underlying relationships within and across Machine Learning algorithms and the paradigms of supervised and un-supervised learning.
4. Utilize the structure and design concepts of neural networks applications to solve real life problems
5. Plan and execute successful machine learning and big data projects, including selecting an adequate process for this specific task and avoiding the machine learning pitfalls.
6. Understand the issues raised by current research in the field of machine learning

6. Detailed Syllabus

UNIT	CONTENTS	Contact Hrs
Unit – I	Review of Statistical Concepts: Mean, Median, Mode, Outliers, Range, Average Deviation, Absolute Deviation, Squared Deviation, Standard Deviation, Total Sum of Squares. Review of Linear Algebra: Vectors and Matrices, Addition and Multiplication of Scalars, Matrix Multiplication Properties, Inverse and Transpose. Introduction to Machine Learning: What is Machine Learning, Introduction to ML's three approaches: Supervised, Unsupervised and Reinforcement Learning. Introduction to Matlab/Octave: Basic Operations, Moving Data Around, Flow Control, Vectorization. Introduction to Python: Basic Operations, Lists, Tuples, Dictionaries, Flow Control, Strings, File handling, Numpy, Scikit-learn, Orange.	10
Unit – II	Validation Techniques: Hold out, K-Fold Cross Validation, Leave one out, Bootstrapping. Supervised Learning Algorithms: Linear Regression, Logistic Regression, Decision Trees, Random Forest, Support Vector Machine, K-	10

	Nearest Neighbours, CN2 Algorithm, Naive Bayes, Artificial Neural Networks. Ensemble Learning: Bagging, Random Forest, AdaBoost, Bucket of Models, Stacking	
Unit – III	Clustering: K-means, Silhouette Scores, Hierarchical Clustering, Fuzzy c-means, DBScan Dimensionality Reduction: Low Variance Filter, High Correlation Filter, Backward Feature Elimination, Forward Feature Selection, Principle Component Analysis, Projection Methods. Association Rule Learning: Support, Confidence, Lift, Conviction, Apriori Algorithm, Eclat Algorithm.	8
Unit – IV	The Rise of Deep Learning: Mask R-CNN, Yolo, AlexNet, VGG, MobileNet, Deeplab, Fully Convolutional Networks, Image captioning (CNN+LSTM), Word2Vec, Doc2Vec, Autoencoder. Deep Learning Tools: TensorFlow, PyTorch, Keras	14
Unit – V	Reinforcement Learning: Agent, Environment, Rewards, States, Actions, Policy, Value, Q-value, Trajectory, Three approaches to Reinforcement Learning, Markov Decision Process, Q Learning, State-Action-Reward-State-Action (SARSA), Deep Q-Network (DQN), Deep Deterministic Policy Gradients (DDPG), Monte Carlo Methods, OpenAI Gym.	7
Total		49

Text and Reference Books

1. "Machine Learning For Dummies", John Paul Mueller and Luca Massaron
2. "A Course in Machine Learning", Hal Daumé III.
3. "Programming Collective Intelligence: Building Smart Web 2.0 Applications", Toby Segaran
4. "Building Machine Learning Systems with Python", WilRichert and Luis Pedro Coelho
5. "Learning scikit-learn: Machine Learning in Python", Raúl Garreta and Guillermo Moncecchi
6. "Machine Learning in Action", Peter Harrington

Course Code: TCS 504			Course Title: System Design		
Credits:	3	Semester:	V	Category of Course:	DS
TEACHING PERIODS (In hours per week)			TW+SL (in hours per semester)	Relative Weight	
L	T	P	45	CIE	MSE ESE
3	0	0		25	25 50
Examination Duration (Hrs):			Theory 3	Practical 0	

Pre-requisite: DBMS (TCS 503), Operating Systems (TCS 502), Computer Networks (TCS 514).

Course Outcome:	After completion of the course, the students will be able to: CO1: Analyze the scalability, database consistency, and communication protocols CO2: Design efficient and secure data handling and asynchronous messaging workflows using NoSQL databases and message queues CO3: Evaluate architectural patterns, rate-limiting algorithms, and distributed caching. CO4: Implement microservice decomposition patterns, service discovery mechanisms, and fault-tolerance strategies CO5: Synthesize large-scale architectures, formulating technical schemas and infrastructure components for real world services or applications
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Details of the Course:

Sl. No.	Contents	Contact Hours
1	Unit 1: The Foundations of Scalability Architecture Basics: Evolution from Monolithic to N-Tier Architecture; Vertical Scaling (Scale-up), Horizontal Scaling (Scale-out); Latency, Throughput (Disk seek, L1 cache ref, network trip). Networking & Communication: Communication Protocols-REST, RPC, GraphQL, WebSockets; Load Balancing - L4 (Transport Layer), L7 (Application Layer), Algorithms- Round Robin, Least Connections, Consistent Hashing.	10

	Caching Strategies: Reading Strategies- Cache-Aside, Read-Through; Writing Strategies- Write-Through, Write-Back, Write-Around; Eviction Policies- LRU, LFU; Tool Focus- Redis/Memcached architecture basics. Database Concepts for Scale: ACID (SQL), BASE (NoSQL) consistency models; CAP Theorem- Understanding Trade-offs (CP vs AP systems); Replication- Master-Slave, Master-Master; Sharding-Horizontal partitioning, Sharding keys, and Hotspot problems.	
2	Unit 2: Data Handling & Asynchronous Processing NoSQL Deep Dive: Types of NoSQL databases and their specific use cases-Key-Value, Document, Columnar, Graph. Asynchronous Messaging & Queues: Synchronous and Asynchronous processing: Message Queues- Decoupling producers and consumers (RabbitMQ), Pub/Sub Models- Fan-out architecture (Kafka); Handling backpressure and dead-letter queues. Identity & Security: Authentication, Authorization; Session-based Auth, Token-based Auth (JWT); OAuth 2.0, OpenID Connect flows; HTTPS/SSL Handshake basics in a distributed context.	9
3	Unit 3: Case Studies Case Study: Design a URL Shortener: Key Challenges- Generating unique IDs, handling heavy read-heavy traffic; Application- Using KGS (Key Generation Service) to pre-generate keys; Bottleneck Analysis-Cache eviction policies for "hot" URLs. Case Study: Design a Rate Limiter: Key Challenges- Preventing DDOS and API abuse; Algorithms- Token Bucket, Leaky Bucket, Sliding Window; Application- Using Redis for distributed counting. Lab/Workshop: Schema Design: Activity- Students must design the Database Schema (SQL vs NoSQL) for a "Pastebin" service and justify their choice based on Unit 1 concepts (CAP theorem).	8
4	Unit 4: Advanced Reliability & Microservices Microservices Architecture: Monolith services and Micro services - Decomposition patterns; Service Discovery- Client-side discovery, Server-side discovery.API Gateway- Rate limiting, routing, and aggregation at the edge. Distributed Transactions & Consistency: Problem of distributed commits; Two-Phase Commit - Risks and blocking; Saga Pattern- Choreography, Orchestration for long-running transactions. Reliability & Observability: Resiliency Patterns- Circuit Breaker, Bulkhead, Retry with Exponential Backoff; Observability- Distributed Tracing, Logging, and Metrics. Content Delivery & Storage: Content Delivery Networks - Push, Pull CDNs; Blob Storage - Storing large media files (S3 structure, multipart uploads).	10
5	Unit 5: Advanced Case Studies Case Study: Design a Social Media Feed (Instagram/Twitter): Key Challenges- Feed generation (Pull vs. Push/Fan-out on write); Application- Hybrid approach for celebrities and normal users; Storage- Storing graph data (Followers) vs. Blob data (Images). Case Study: Design a Ride-Sharing Service (Uber/Lyft): Key Challenges- Real-time location updates, Geospatial databases (QuadTrees vs. Geohashes); Application: Matching drivers to riders efficiently; Consistency - Handling "Double Booking" using distributed locks; Case Study: Design a Chat Application (WhatsApp/Discord): Key Challenges- Real-time delivery, presence indicators (Online/Offline); Application- WebSocket connections, preserving message order, historical chat storage (Cassandra/HBase).	8